Questions

Q1.

Figure 3 is a velocity / time graph for 15 s of a cyclist's journey.

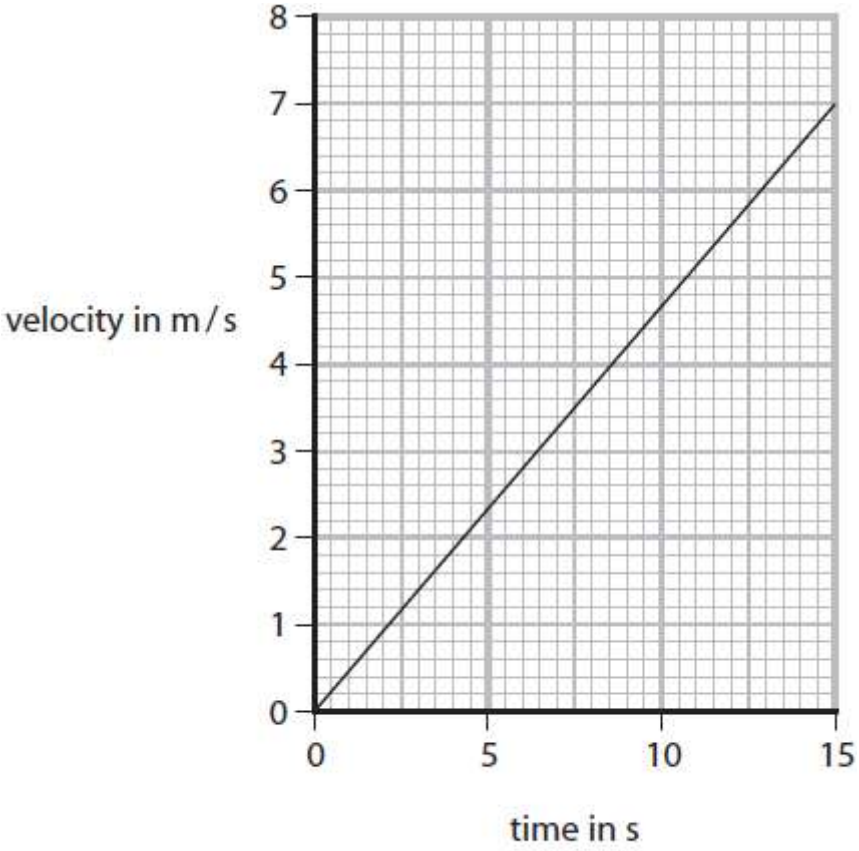


Figure 3

(i) Calculate the distance the cyclist travels in the 15 s.

(3)

distance = ... m

(ii) Another cyclist starts from rest, but his acceleration decreases as time increases.
Sketch the velocity / time graph for this cyclist on Figure 3.

(2)

(Total for question = 5 marks)

Q2.

The graph in Figure 1 shows how the braking distance, *d*, of a car depends on the velocity, *v*, of the car when the brakes are first applied.

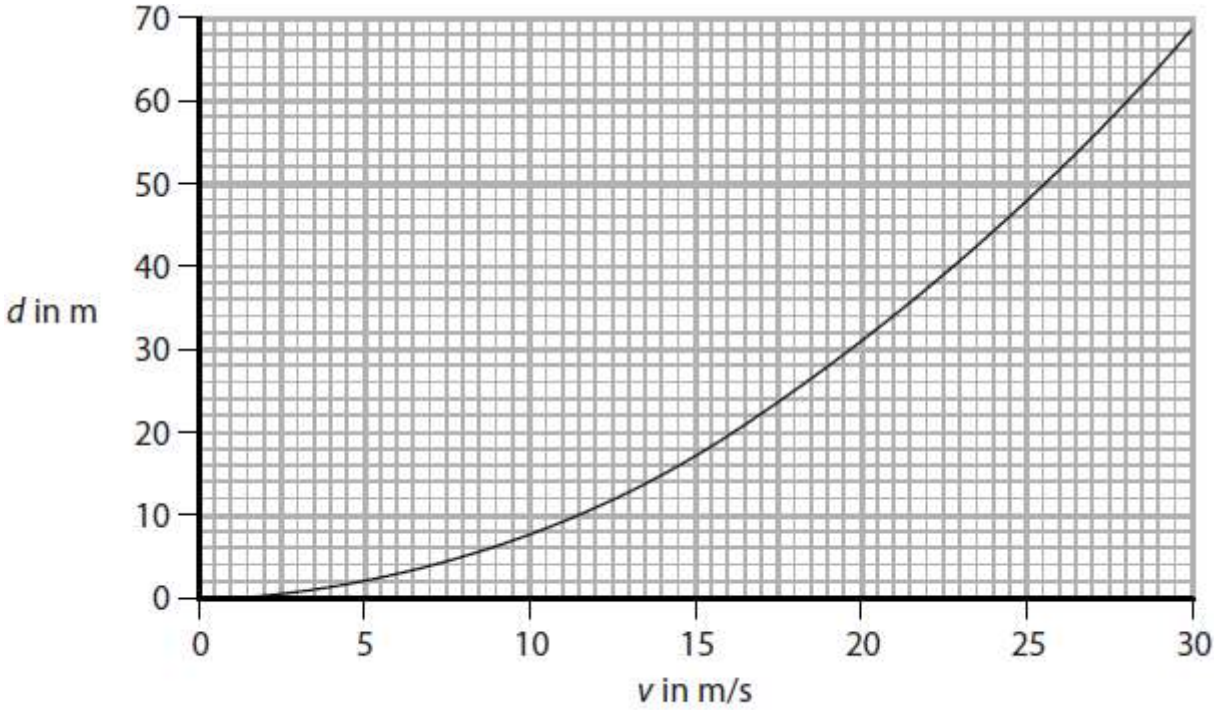


Figure 1

An equation relating braking distance, *d*, to velocity, *v*, is

$$d = \frac{v^2}{C}$$

where *C* is a constant.

Use the equation and data from the graph in Figure 1 to calculate a value for *C*.

Give a unit for *C*.

(4)

C =..... unit .....

(Total for question = 4 marks)

Q3.

The symbol '*g*' can be used to refer to the acceleration due to gravity.

The acceleration due to gravity '*g*' has the unit of m/s^2.

'*g*' can also have another unit.

Which of these is also a unit for *g*?

(1)

- ☐ **A** J/kg
- ☐ **B** J/kg^2
- ☐ **C** N/kg
- ☐ **D** N/kg^2

(Total for question = 1 mark)

Q4.

Figure 11 shows athletes running at a constant speed on a curved section of a track.

(Source: © Photo and Co/Getty Images)

Figure 11

(i) Even though they are running at a constant speed, the velocity of each athlete is changing.
State why the velocity of each athlete is changing.

(1)

.....

.....

(ii) The athletes start the race on a straight section of track.
Figure 12 shows a distance / time graph for one of the athletes.

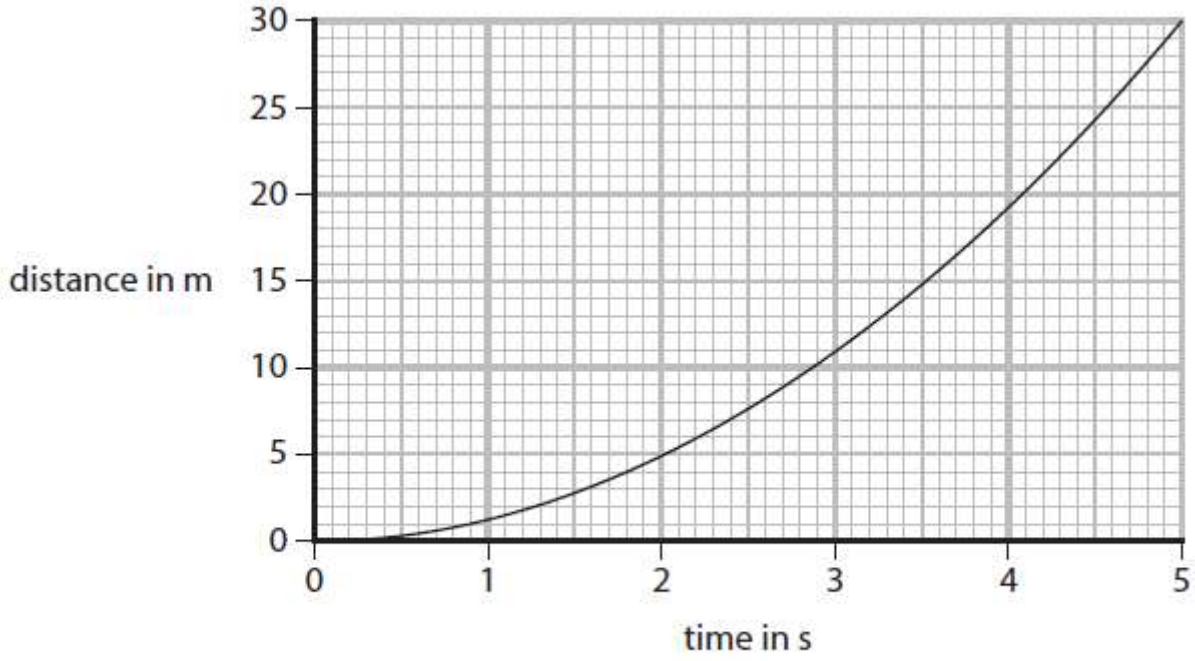


Figure 12

Determine the gradient of this graph at a time of 2.5 s.

(2)

gradient = ... m/s

(Total for question = 3 marks)

Q5.

Figure 8 is a velocity/time graph for a lift moving upwards in a tall building.

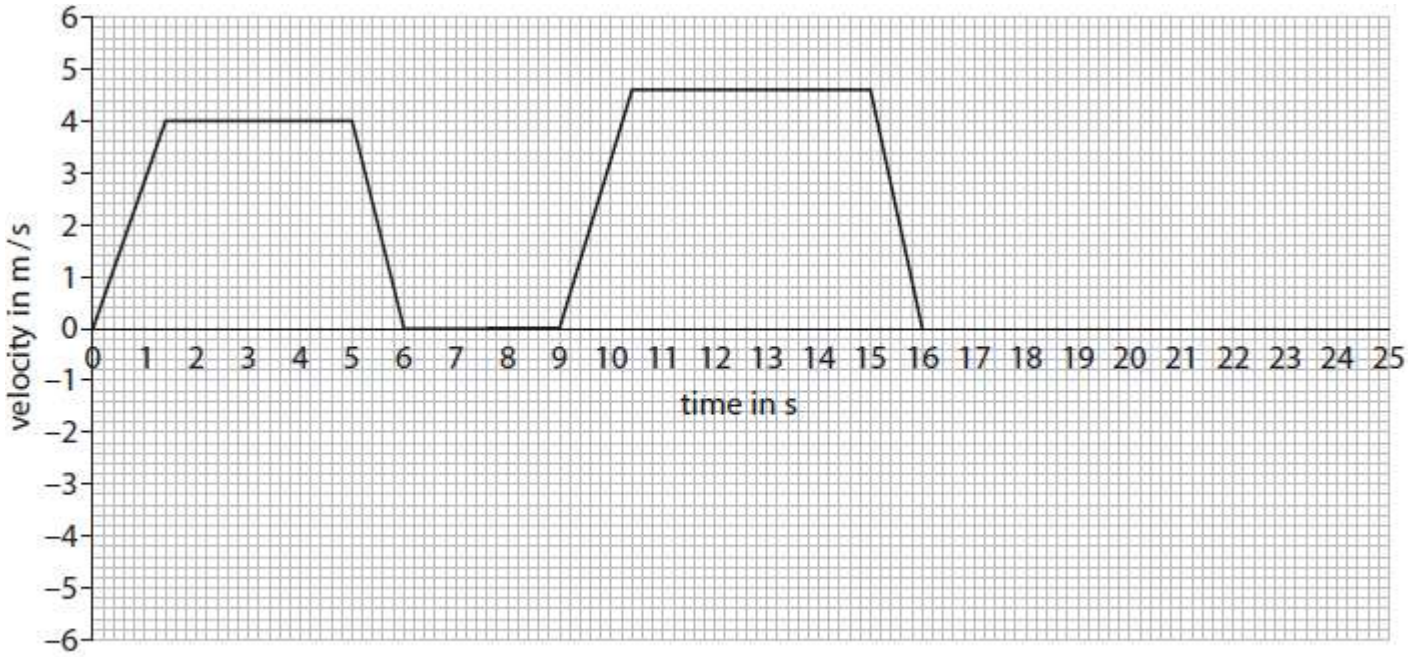


Figure 8

Use the graph in Figure 8 to determine the maximum velocity of the lift during the first 16 s.

(1)

maximum velocity = ... m/s

(Total for question = 1 mark)

Q6.

(i) State the equation that relates acceleration to change in velocity and time taken.

(1)

(ii) A van accelerates from a velocity of 2 m / s to a velocity of 20 m / s in 12 s.
Calculate the acceleration of the van.

(2)

acceleration = ... m / s^2

(Total for question = 3 marks)

Q7.

Two people, L and M, have a 100 m race.

L starts running before M.

Figure 4 shows a distance/time graph of the race.

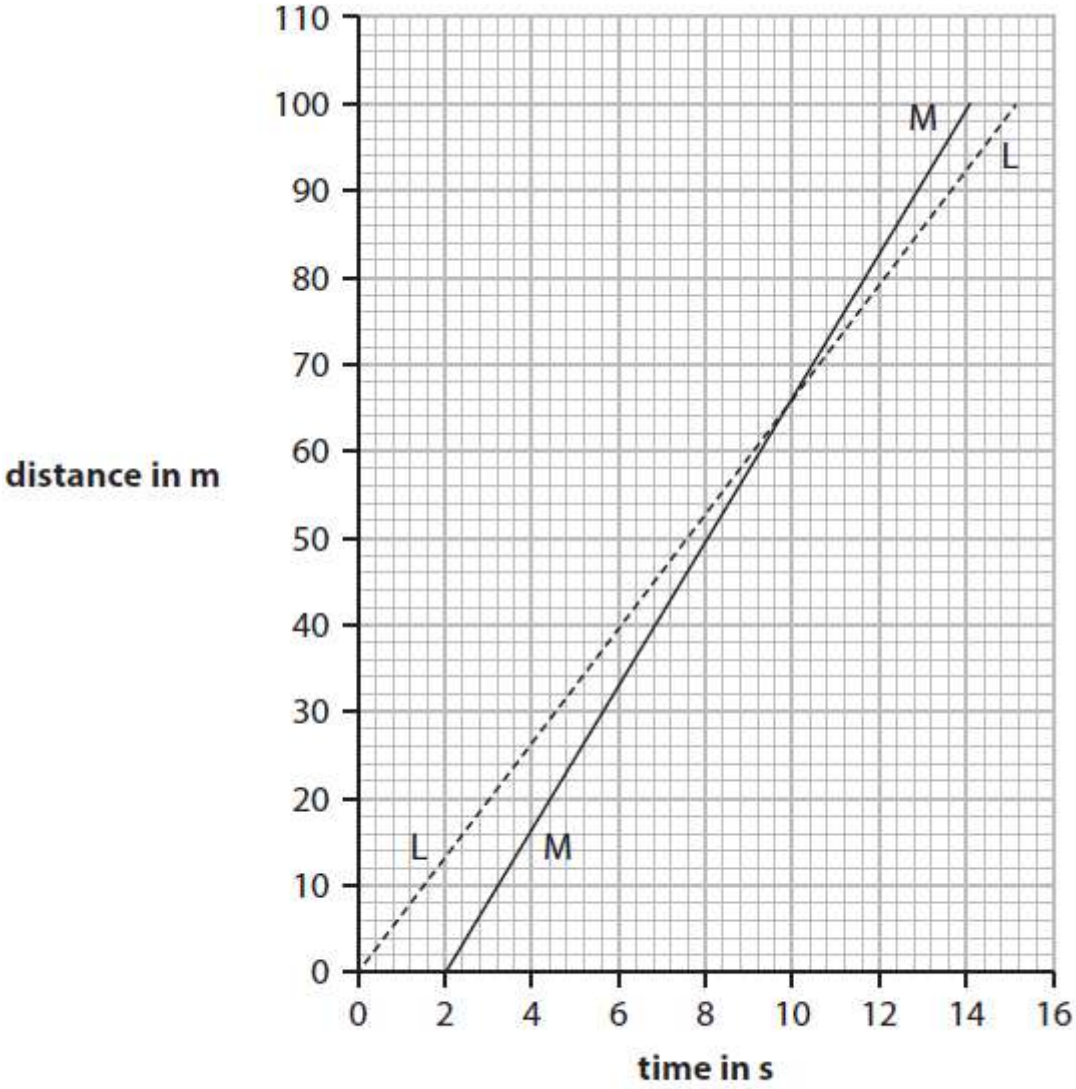


Figure 4

(i) State the **distance** that L has run when M overtakes.

(1)

distance = ... m

(ii) Calculate the velocity of L when running the 100 m race.

(2)

velocity = ... m/s

(Total for question = 3 marks)

Q8.

Figure 11 shows an object moving in a circular path.

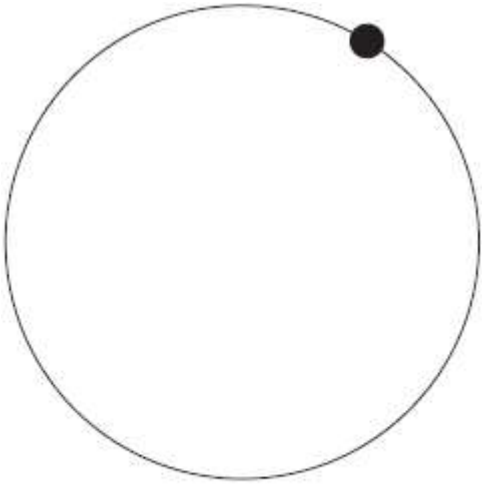


Figure 11

(i) Draw an arrow on Figure 11 to show the direction of the force that keeps the object moving in a circular path.

(1)

(ii) The object in Figure 11 is moving at constant speed.
Explain why it is not moving with constant velocity.

(2)

.....

.....

.....

.....

(Total for question = 3 marks)

Q9.

(i) An aircraft starts from rest and accelerates along the runway for 36 s to reach take-off velocity.

Take-off velocity for this aircraft is 82 m/s.

Show that the acceleration of the aircraft along the runway is about 2 m/s^2 .

Assume the acceleration is constant.

(2)

(ii) Calculate the distance the aircraft travels along the runway before take-off.

Use the equation

$$v^2 - u^2 = 2ax$$

(3)

distance = ... m

(iii) Suggest **one** reason why the length of the runway used is always much longer than the calculated distance that the aircraft travels along the runway before take-off.

(1)

.....

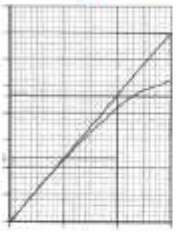
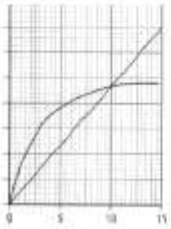
.....

(Total for question = 6 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i) CLIP WITH GRAPH	distance = area under graph (1) ½ × 7 × 15 (1) 52(.5) (m) (1)	attempt to find area seen on graph correct area(s) identified including calculation 53 (m) allow 7 × 15 or 105 for 1 mark only award full marks for the correct answer with no working	**(3)**

Question number	Answer	Additional guidance	Mark
(ii) CLIP WITH GRAPH H paper	(curve) <u>starting</u> from 0,0 (1) of decreasing gradient (1)	curve can be above or below the line both of these are acceptable  	**(2)**

Q2.

Question Number	Answer	Additional guidance	Mark
	use values from graph (1) e.g. v = 20, d = 31 rearrangement (1) $C = \frac{v^2}{d}$ evaluation (1) (C=) 13 unit (1) m/s^2	accepting values to within one square allow numbers from 12.5 to 13.5 award 3 marks for the correct numerical answer without working independent mark	**(4)**

Q3.

Question Number	Answer	Mark
	C N/kg **is the only correct answer** *A J/kg is not dimensionally the same as m/s^2* *B J/kg^2 is not dimensionally the same as m/s^2* *D N/kg^2 is not dimensionally the same as m/s^2*	**(1)** AO 1 1

Q4.

Question number	Answer	Additional guidance	Mark
(i)	Any **one** from: direction is changing (1) velocity is a vector (1)	 allow velocity has magnitude and direction ignore acceleration ignore reference to centripetal force ignore speed is changing	**(1)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	tangent drawn to curve (at 2.5 s) (1) evaluation of gradient (1) (gradient =) 6.0 ± 1.0 (m/s)	accept hypotenuse of triangle dependent on MP1	**(2)** **AO3.2**

Q5.

	Answer	Additional guidance	Mark
	4.6 (m/s)	allow any value between 4.5 and 4.7 (m/s) inclusive	**(1)** **AO3**

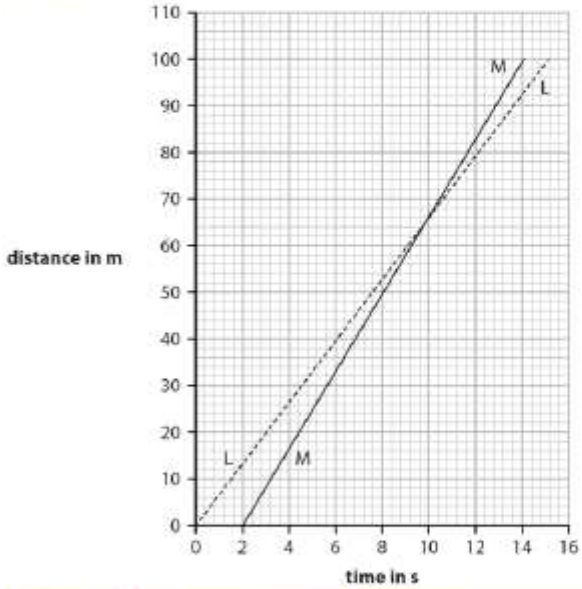
Q6.

Question number	Answer	Additional guidance	Mark
(i) CLIP WITH ii	acceleration = $\frac{\text{change in velocity}}{\text{time (taken)}}$	$a = \frac{v-u}{t}$ $a = \frac{\Delta v}{t}$ $\frac{v}{t}$ allow correct rearrangements seen here or in ii	**(1) grad**

Question number	Answer	Additional guidance	Mark
(ii) CLIP WITH i	substitution (1) $\frac{20 - 2}{12}$ evaluation (1) 1.5 (m/s²)	 $\frac{18}{12}$ -1.5 (m/s²) award full marks (1 in i and 2 in ii) for the correct answer without working, award 1 mark if 20-2 or 18 or 2-20 is seen and no other marks are scored If (incorrectly) $a = \frac{v^2 - u^2}{t}$ given in 3bi $a = \frac{20^2 - 2^2}{12}$ OR = 33 scores 1 mark	**(2)**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	66 (m)	allow values between 64 and 68 inclusive	1 AO3.2
		allow values between 32 and 36 inclusive as the distance L has to run after M overtakes in this context	

Question number	Answer	Additional guidance	Mark
(ii)	select (1) $v = \frac{x}{t}$	allow any identifiable distance from graph divided by any identifiable time from graph e.g. $\frac{100}{15.2}$  	2 AO2.1
	evaluation (1) 6.6 (m/s)	allow values that round to between 6.5 (m/s) and 6.7 (m/s) for example 6.666 (m/s) or 6.579 (m/s)	

		award full marks for correct answer without working	

Q8.

Question Number	Answer	Additional guidance	Mark
(i)	single arrow towards centre of the circle applied to the object (1)	judge by eye	**(1)**

Question Number	Answer	Additional Guidance	Mark
(ii)	an explanation including velocity is a vector (1) (because) direction changes (1)	 velocity has (magnitude and) direction / velocity is speed in a (certain) direction	**(2)**

Q9.

	Answer	Additional guidance	Mark
(i)	selection and substitution (1) (a =) $\frac{82\ (-\ 0)}{36}$ evaluation (1) 2.3 (m/s^2)	**note: this is a "show that" question** accept any value that rounds to 2.3 (m/s^2) accept 2.2 (m/s^2) for 1 mark maximum answer of 2 (m/s^2) without a substitution scores 0 marks	**(2)** **AO2**

	Answer	Additional guidance	Mark
(ii)	substitution (1) 82^2 (– 0^2) = 2 x 2.3 x *x* rearrangement (1) (x) = $\frac{82^2\ (-\ 0^2)}{2 \times 2.3}$ evaluation (1) 1500 (m)	allow substitution and rearrangement in either order accept 2, 2.2, 2.27, 2.3 for "a" throughout (x) = $\frac{v^2\ (-\ u^2)}{2 \times a}$ ignore sign accept 1460 (m) allow answers in the ranges: 1460 (m) to 1482 (m) 1520 (m) to 1530 (m) 1680 (m) to 1700 (m) award full marks for correct answer without working	**(3)** **AO2**

	Answer	Additional guidance	Mark
(iii)	one statement from take off aborted (1) mechanical/engine failure (1) acceleration reduced (1) weather related reasons (1) larger mass / heavier plane / extra passengers (1) (longer runway required) for landing (1)	any other sensible suggestion	(1) AO3